

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Tech. Examination (CL)

April 2013

CL 405 Estimation, Costing & Financial Concepts -II

Date: 24.04.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 Figure 1 & Table-1. Shows the details of a Roof shed. Workout the total quantity of the [07]
(a) steel work for truss. The effective span of truss is 12m and spacing of truss is 3.0m. The inside length of shed is 15m. Assume suitable data if required.
- Q - 2 Workout the quantity of foundation (up to plinth level) work for drawings given in [14]
(a) Figure 2.

OR

- Q - 2 Workout the quantity of following items for first floor, drawings given in Figure 2. [14]
(a) (i) Inside plaster work. (ii) Flooring (iii) Masonry work (iv) R.C.C work (Beam, Column and slab only)
- Q - 3 Give general guide lines for report preparation of any civil work. Also include the [05]
(a) Process of documentation of report.
- (b) Workout quantity of septic tank shown in Figure 3 [09]

OR

- Q - 3 List the document required for the project report preparation of road work. [05]
(a)
- (b) Workout the quantity of sanitation work from Figure 2. [09]

SECTION - II

- Q - 4 Work out quantity of following items for village road culvert shown in Figure 4. [07]
(a) (i) Excavation (ii) Cement concrete 1:3:6 (iii) Brick work in 1:4 (iv) R.C.C work in 1:2:4
- Q - 5 Define the following. [07]
(a) a) Sinking fund b) Depreciation c) Salvage value d) Out-goings e) Years purchase f) Scrap value. g) Capitalized value
- (b) The present value of a property is Rs. 1,15,000 out of which the cost of land is Rs. 25,000. [07]
The owner of the property expects 7.5% return on the cost of construction and 6.5% return on the cost of land. If the future life of the building is estimated as 80 years and at the end of its useful life, Rs. 1,35,000 will be required for replacing the construction.
Calculate the standard rent of property assuming
1. Rate of interest for sinking fund is 5%
 2. Annual repairs cost 1% of the cost of construction.
 3. All other outgoing taxes shall be 30% of the net annual income of the property.
 4. The scrap value of building at the expiry of its useful life is estimated as 10% of its present value.

OR

Q - 5 Write short notes on Mortgage

[07]

(a)

- (b) A three storied building is standing on a plot of land measuring 800 sqm. The plinth area of the each storey is 400sqm. The building of R.C.C framed structure and the future life may be taken as 70 years. The building fetches a gross rent of Rs. 1500. Per month. Work out the capitalized value of the property on the basis of 6% net yield. For sinking fund 3% compound interest may be assumed. Cost of land may be taken Rs. 400per sqm. Other data required may be assumed suitably. Gross income per year = $1500 \times 12 = \text{Rs}18000$.

Q - 6 Distinguish between Secured advance and running account bill.

[04]

(a)

- (b) Reduced levels of ground along the centre line of a proposed road from chainage 0 to 200m are given below. The formation level at 40 m chainage is 102.75. The formation of road from 0 to 80 has rising gradient 1 in 40. The formation level has falling gradient 1 in 100 from 80 to 200m chainage. The formation width of road is 12m and side slope is 2:1.

Chainage (m)	0	20	40	60	80	100	120	140	160	180	200
G.L	101.5	100.9	101.5	102	102.85	101.65	101.95	100.7	101.25	99.9	100.6

OR

Q - 6 State the condition of release of first and final bill of work.

[04]

(a)

- (b) Prepare estimate for the following items for construction of a new state highway having one kilometer length. The formation width of road is 10 meter, average height of bank is 1 meter and side slop is 2:1. The metaled width is 3.80m and three coats of metalling are to be provided as per cross section. The surface shall be finished with two coats of painting. (i) Surveying (ii) land acquisition (iii) Metaling (iv) Earth work and Plantation (v) Painting or Black top surfacing.

Table 1.0 Weight of steel section(Q.1)

Section Size	Wt/unit	Section Size	Wt/unit
2.75 x 50 x 8mm angle	7.4kg/m	50 x 8mm F.B	3.1kg/m
75 x 50 x 10mm angle	9.0kg/m	80 x 6mm F.B	3.8kg/m
60 x 45 x 8mm angle	6.4kg/m	32 x 6mm F.B	1.5kg/m
50 x 50 x 6mm angle	4.5kg/m	Gusset plate Size	Wt/sq m
75 x 75 x 6mm angle	8.9kg/m	8mm	62.8kg/sqm
75 x 75 x 10mm angle	11kg/m	Base plate Size	Wt/sq m
125 x 75 x 8mm angle	12.1kg/m	12mm	94.2kg/sqm

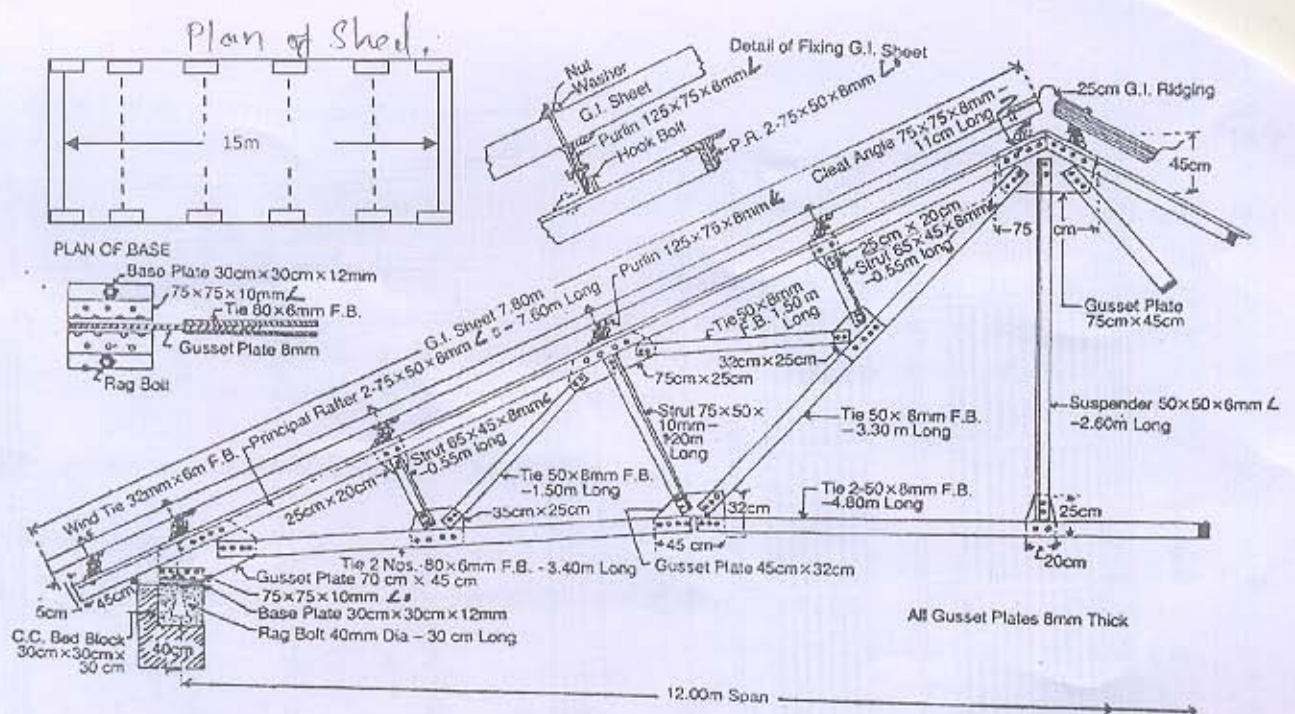


Figure 1 Q.1

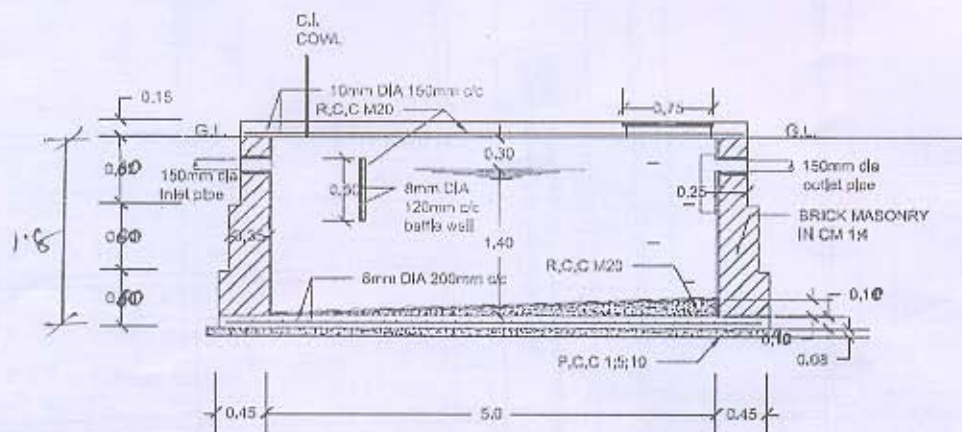
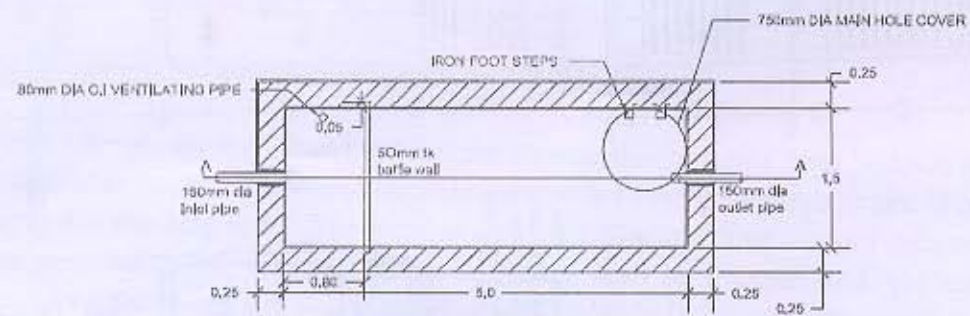


Figure 3 Q.3(b)

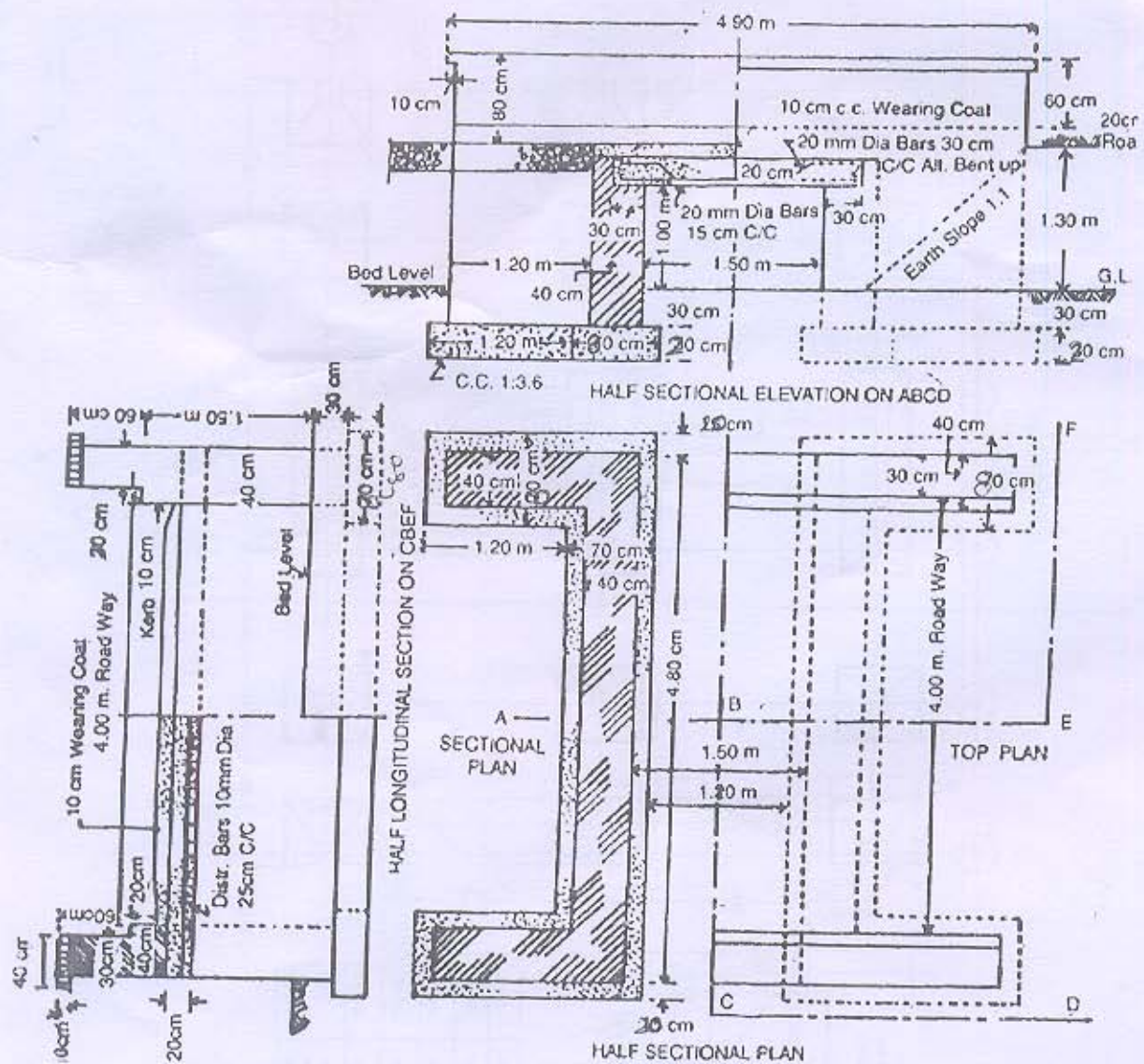


Figure 4 Q.4 (a)

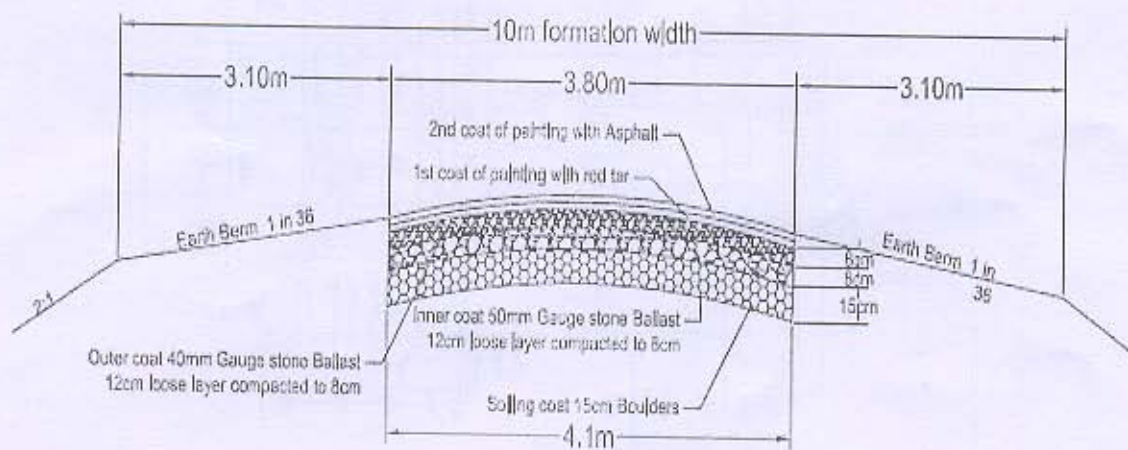


Figure 5 Q.6 (b)